

CSC4402: Introduction to Database Management Systems

Fall – 2026

Assignment - 1 (10 Points)

Logistics:

Deadline: 09/18/2026 11:59 PM CT

Deliverable: A single PDF document containing the answers to all questions

Submission: [Gradescope](#)

Questions (Correct answers to each question will amount to 2.5 points (4 x 2.5 = 10)):

1. **Question 1:** Consider the following bank database and employee database:

branch (*branch_name*, *branch_city*, *assets*)

customer (*ID*, *customer_name*, *customer_street*, *customer_city*)

loan (*loan_number*, *branch_name*, *amount*)

borrower (*ID*, *loan_number*)

account (*account_number*, *branch_name*, *balance*)

depositor (*ID*, *account_number*)

- (a) Identify the primary key for all relations in the given database (1 point)
 - (b) **Consider the employee database from Question 5.** Write a relational algebra operation to find and show the names of all employees who live in London and have a salary of less than 70,000 (1.5 points).
2. **Question 2:** Consider the bank database from Question 1. Draw the schema diagram of the given database (2.5 points).
 3. **Question 3:** Consider the university database (See the data packet attached with the assignment).
 - (a) Write an SQL query to increase the salary of the lowest-paid computer science instructor by 8% (1 point).
 - (b) Write an SQL query to find the names and average salaries of all departments whose average salary is between \$52000 and \$75000 (1.5 points).

4. **Question 4:** Consider the following employee database.

employee (ID, *person_name*, *city*)

works (ID, *company_name*, *salary*)

company (company_name, *city*)

manages (ID, *manager_id*)

- (a) Write an SQL query to find the ID of each employee who earns more than every employee of the “Bank of America” (1 point).
- (b) Consider the following query: “Give each manager of “JP Morgan Chase” a 9% raise capped at \$100,000. If the salary becomes greater than \$100,000, give them a 3% raise”. The update can be performed using the following SQL query:

```
UPDATE works T
SET T.salary = T.salary * 1.03
WHERE T.ID IN (SELECT manager.id FROM manager)
      AND T.salary * 1.09 > 100000
      AND T.company_name = “JP Morgan Chase”;
```

```
UPDATE works T
SET T.salary = T.salary * 1.09
WHERE T.ID IN (SELECT manager.id FROM manager)
      AND T.salary * 1.09 <= 100000
      AND T.company_name = “JP Morgan Chase”;
```

While correct, it might be unsafe to write this query to perform the update. Why (0.5 points)? Write an alternative safe but equivalent SQL query (1 point).